Simon Kaplinsky

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EDUCATION & HONORS

Carnegie Mellon University Pittsburgh, PA
B.S. Electrical and Computer Engineering ***Expected May 2029***
GPA: 3.87 / 4.00 | Dean's List (Spring 2026)
Relevant Coursework: Electronic Devices and Analog Circuits |Structure and Design of Digital Systems | Fundamentals of Electromagnetics | Signals and Systems | PCB Fabrication

EXPERIENCE

Electrical Team Member Sep 2025 – Present
CMU Rocket Command *Pittsburgh, PA*

- Designing, laying out, and debugging custom circuits in KiCad under tight space and weight constraints, maintaining reliability through launch, flight, and landing loads
- Build an ESP32-based payload computer for NASA's University Student Launch Initiative, integrating an altimeter, conductivity sensor, motor driver, and SD-card data logging
- Write payload firmware that autonomously detects deployment and landing, then drives a motor to collect and measure a soil sample
- Perform board-level bring-up and fault isolation across repeated ground and flight tests
- Prepare flight readiness and performance documentation for submission to NASA's University Student Launch Initiative (USLI)

Teaching Assistant — Electronic Devices and Analog Circuits (18-220) Aug 2026 – Present
Carnegie Mellon University, Department of ECE *Pittsburgh, PA*

- Coach students through lab and homework debugging, guiding fault isolation in analog circuits and reasoning from measurement back to device behavior
- Lead small-group review sessions on core course topics and host weekly office hours

Software Engineering Summer Associate May 2026 – Aug 2026
Byline Travel, Inc. *Palo Alto, CA*

- Developed production Python and TypeScript for an AI-powered travel platform and optimized PostgreSQL-backed services, merging 40 pull requests over a 14-week internship
- Engineered an agentic web scraper to autonomously gather and structure point-of-interest data across 15 sources while getting around bot detection

PROJECTS

Interactive Wearable Costume June 2026 – October 2026
Personal Project *Pittsburgh, PA*

- Designed a sensor-to-actuator signal chain mapping heart rate, gyroscope, and capacitive touch input to addressable LED drivers in real time; hand-wired and soldered all circuitry and wrote the device firmware

SKILLS

Circuit Design & PCB Development: Schematic capture, PCB layout and fabrication, analog/digital circuit analysis, sensor integration, board bring-up
Digital Design: RTL design, Datapaths and ALUs, finite state machines, CPU architecture, CMOS modeling
Tools & Platforms: KiCad, AMD Vivado, ESP32, Arduino, oscilloscopes, Git/GitHub, PostgreSQL, 3D printing
Programming: C, C++, Python, TypeScript

ACTIVITIES & LEADERSHIP

Social Chair, Mr. Yuk Ultimate Frisbee May 2026 – Present
Carnegie Mellon University *Pittsburgh, PA*

- Recruited 80+ prospective players through campus outreach to fill a 24-person competitive roster; manage team outreach and run official university events during orientation week